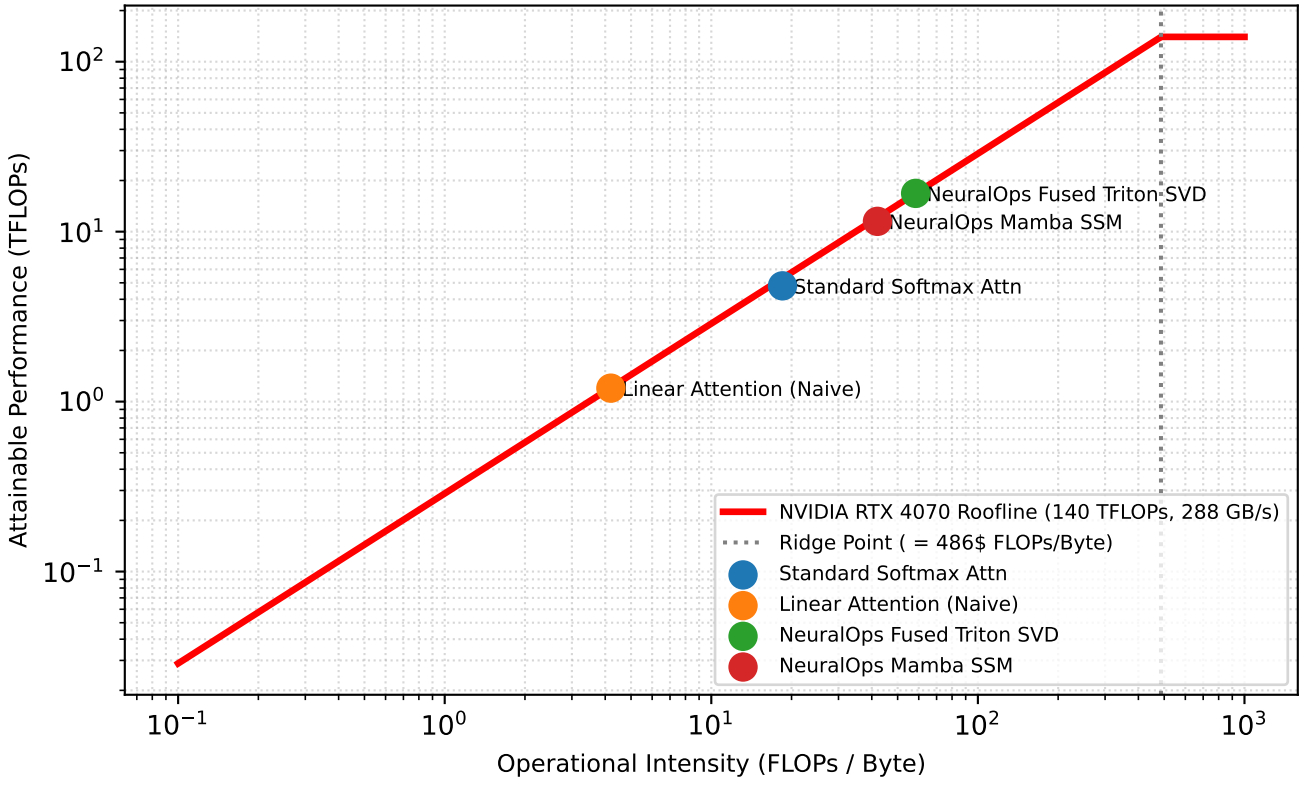
Hardware Roofline & Operational Intensity Comparison
Attainable Performance (TFLOPs)
Operational Intensity (FLOPs / Byte)
10^{2}
10^{1}
10^{0}
10^{-1}
10^{-1}
10^{0}
10^{1}
10^{2}
10^{3}
NeuralOps Fused Triton SVD
NeuralOps Mamba SSM
Standard Softmax Attn
Linear Attention (Naive)
NVIDIA RTX 4070 Roofline (140 TFLOPs, 288 GB/s)
Ridge Point (= 486$ FLOPs/Byte)
Standard Softmax Attn
Linear Attention (Naive)
NeuralOps Fused Triton SVD
NeuralOps Mamba SSM